

Maximum Cardinality of Divisibility-Free Subsets in Rows of Pascal's Triangle: OEIS A399060

Felix Huber
Maihofhalde 20
6006 Luzern, Switzerland
`felix.68@gmx.ch`
ORCID: 0009-0005-1568-1579

Abstract

We study divisibility-free subsets of the distinct entries in a row of Pascal's triangle. Let $a(n)$ be the maximum number of entries in the n th row such that none divides another. The aim of this paper is to investigate the asymptotic behavior of $a(n)$ and to derive general lower bounds.

We first characterize divisibility between neighboring binomial coefficients in the same row and reduce the general divisibility problem to products of consecutive integers. A theorem of Erdős and Straus then shows that primes between the positions of two coefficients act asymptotically as barriers to divisibility. From this observation we construct large divisibility-free sets and obtain $a(n) \geq (1 - o(1))n / \log n$. For rows of the form $n = p - 1$, where p is prime, we may also use the congruence $\binom{p-1}{k} \equiv (-1)^k \pmod{p}$. It follows that comparable coefficients in a substantial index range must be separated by at least a logarithmic distance. Combined with a result of Bazzanella, Languasco, and Zaccagnini on logarithmic prime gaps, this yields the stronger bound $a(p - 1) \geq (c - o(1))p$ with an absolute constant $c > 0$. Finally, we show that $a(n)$ can be computed exactly using Dilworth's theorem together with a maximum matching.

Keywords: Pascal triangle, binomial coefficient, divisibility, antichain, prime gap, integer sequence.

2020 Mathematics Subject Classification: Primary 11B65; Secondary 05D05.

1 Introduction

The entries of a row of Pascal's triangle have a rich arithmetic structure. [6, 4] In this paper we consider the following extremal problem:

How many distinct entries of a row of Pascal's triangle can be selected such that none of the selected entries divides another?

For $n \geq 0$, let

$$P_n = \left\{ \binom{n}{k} : 0 \leq k \leq \left\lfloor \frac{n}{2} \right\rfloor \right\}$$

be the set of distinct entries in the n th row of Pascal's triangle. By symmetry, it is enough to consider the first half of the row.

We denote by $a(n)$ the maximum size of a subset of P_n whose elements are pairwise incomparable with respect to divisibility. In other words, we want to select as many distinct numbers as possible from a Pascal row without having a divisibility relation between any two selected numbers.

Even small examples show that different rows can behave quite differently.

Example 1. For $n = 6$ the row is

$$1, 6, 15, 20, 15, 6, 1.$$

The distinct entries are 1, 6, 15, 20. The numbers 6, 15, 20 are pairwise incomparable under divisibility. On the other hand, 1 cannot be chosen together with any other number, since 1 divides every other entry. Therefore

$$a(6) = 3.$$

Example 2. For $n = 11$ we obtain

$$1, 11, 55, 165, 330, 462.$$

Here we have the divisibility chain

$$1 \mid 11 \mid 55 \mid 165 \mid 330.$$

A divisibility-free set can contain at most one element from this chain. Since 330 and 462 do not divide each other,

$$a(11) = 2.$$

The sequence $a(n)$ is listed in the OEIS as [A399060](#) [5] and begins

$$1, 1, 1, 1, 2, 1, 3, 2, 3, 3, 4, 2, 4, 4, 4, 4, 7, 5, 6, 5, \dots$$

This leads to two natural questions:

- When are two binomial coefficients from the same row comparable under divisibility?
- How can this structure be used to construct large divisibility-free sets?

Primes between the positions of two coefficients act asymptotically as barriers to divisibility. This yields the following two results.

Theorem 1. As $n \rightarrow \infty$,

$$a(n) \geq (1 - o(1)) \frac{n}{\log n}.$$

For the special family $n = p - 1$ with p prime, we obtain a linear lower bound.

Theorem 2. *There exists an absolute constant $c > 0$ such that*

$$a(p-1) \geq (c - o(1))p$$

as $p \rightarrow \infty$ through the primes.

The mathematical ideas and proof strategies presented in this paper were developed as part of an AI-assisted research process using ChatGPT by OpenAI. The individual ideas proposed by ChatGPT were reviewed by the author. Further details on the role of AI are provided in the statement at the end of the paper.

2 Neighboring entries: the simplest form of divisibility

Before studying the general divisibility structure of a row of Pascal's triangle, we consider two neighboring binomial coefficients. Their divisibility can be described completely by an elementary condition.

The elementary analysis of divisibility relations between neighboring binomial coefficients, and the resulting investigation of the structure of Pascal rows, were proposed by ChatGPT as the starting point of the proof approach.

Proposition 3. *For*

$$0 \leq k < \left\lfloor \frac{n}{2} \right\rfloor,$$

we have

$$\binom{n}{k} \mid \binom{n}{k+1} \iff k+1 \mid n+1.$$

Proof. The quotient of two neighboring binomial coefficients is

$$\frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}.$$

Since

$$n-k = (n+1) - (k+1),$$

we obtain

$$\frac{n-k}{k+1} = \frac{n+1}{k+1} - 1.$$

Thus the quotient is an integer if and only if $(n+1)/(k+1)$ is an integer, which is equivalent to $k+1 \mid n+1$. $\square$

Example. For $n = 11$ we have $n+1 = 12$. The numbers 1, 2, 3, 4 divide 12, so Proposition 3 gives

$$\binom{11}{0} \mid \binom{11}{1} \mid \binom{11}{2} \mid \binom{11}{3} \mid \binom{11}{4},$$

that is,

$$1 \mid 11 \mid 55 \mid 165 \mid 330.$$

Thus the divisor structure of $n+1$ already controls some of the divisibility relations in row n .

2.1 A first upper bound for $a(n)$

Let $\tau(r)$ denote the number of positive divisors of r .

Corollary 4. *For $m \geq 1$,*

$$a(2m) \leq m + 2 - \tau(2m + 1)$$

and

$$a(2m + 1) \leq m + 3 - \tau(2m + 2).$$

Proof. First let $n = 2m$. The set P_{2m} contains exactly $m + 1$ elements. Every divisor d of $2m + 1$ with $1 \leq d \leq m$ gives, by Proposition 3,

$$\binom{2m}{d-1} \mid \binom{2m}{d}.$$

Among the $\tau(2m + 1)$ positive divisors, only $2m + 1$ itself lies outside $1, \dots, m$. Indeed, $2m + 1$ is odd, so every proper divisor is at most $(2m + 1)/3$, hence at most m . Thus there are exactly $\tau(2m + 1) - 1$ neighboring divisibility relations.

Initially, each element of P_{2m} forms a chain by itself. Since these relations are edges of a path on the ordered positions, every such edge reduces the number of connected chain components by one. Hence we obtain

$$(m + 1) - (\tau(2m + 1) - 1) = m + 2 - \tau(2m + 1)$$

chains. A divisibility-free set can contain at most one element from each chain, so

$$a(2m) \leq m + 2 - \tau(2m + 1).$$

For $n = 2m + 1$ the argument is analogous. The set P_{2m+1} again has $m + 1$ elements. Among the $\tau(2m + 2)$ positive divisors of $2m + 2$, exactly $m + 1$ and $2m + 2$ lie outside the relevant range $1, \dots, m$. Therefore there are $\tau(2m + 2) - 2$ neighboring divisibility relations, giving

$$(m + 1) - (\tau(2m + 2) - 2) = m + 3 - \tau(2m + 2)$$

chains and hence

$$a(2m + 1) \leq m + 3 - \tau(2m + 2).$$

□

This corollary uses only neighboring divisibility. Additional relations between entries farther apart can improve the upper bound.

2.2 Why neighboring relations are not enough

This is already visible in the row $n = 14$:

$$P_{14} = \{1, 14, 91, 364, 1001, 2002, 3003, 3432\}.$$

Since $n + 1 = 15$, the divisors 1, 3, 5 give

$$1 \mid 14, \quad 91 \mid 364, \quad 1001 \mid 2002.$$

Thus the eight entries initially split into five chains:

$$\{1, 14\}, \quad \{91, 364\}, \quad \{1001, 2002\}, \quad \{3003\}, \quad \{3432\}.$$

Corollary 4 gives $a(14) \leq 5$. However,

$$14 \mid 364, \quad 91 \mid 1001,$$

so these chains can be merged further:

$$\{1, 14, 364\}, \quad \{91, 1001, 2002\}, \quad \{3003\}, \quad \{3432\}.$$

Hence $a(14) \leq 4$. On the other hand,

$$\{364, 2002, 3003, 3432\}$$

is divisibility-free, and therefore

$$a(14) = 4.$$

Thus neighboring relations alone do not determine $a(n)$; divisibility between non-neighboring entries must also be considered.

2.3 The extreme case $a(n) = 1$

Proposition 5. *We have*

$$a(n) = 1$$

if and only if

$$n \in \{0, 1, 2, 3, 5\}.$$

Proof. For $n = 0, 1, 2, 3, 5$ the statement is immediate.

Conversely, suppose $a(n) = 1$ and $n \geq 2$. Then every two distinct elements of P_n are comparable under divisibility. The binomial coefficients

$$\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{\lfloor n/2 \rfloor}$$

increase strictly up to the middle of the row. Hence comparability of neighboring terms can only occur from left to right. Thus, for every $0 \leq k < \lfloor n/2 \rfloor$,

$$\binom{n}{k} \mid \binom{n}{k+1}.$$

By Proposition 3,

$$1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor$$

all divide $n + 1$.

If $n = 2m$, then $m \mid 2m + 1$. Since also $m \mid 2m$, we obtain $m \mid 1$, so $m = 1$ and $n = 2$.

If $n = 2m + 1$, then $m \mid 2m + 2$. Together with $m \mid 2m$, this gives $m \mid 2$. Hence $m \in \{1, 2\}$ and $n \in \{3, 5\}$. $\square$

3 Entries farther apart: the hidden product structure

For neighboring entries, the quotient formula from Proposition 3 was sufficient. For positions farther apart, the situation is less transparent. The key step is to translate divisibility between two binomial coefficients into a problem about products of consecutive integers.

The reduction of divisibility between two binomial coefficients to divisibility between two products of consecutive integers was proposed by ChatGPT. This reduction forms the basis for the subsequent application of the result of Erdős and Straus.

For $x \geq 0$ and $d \geq 1$, define

$$A(x, d) = (x + 1)(x + 2) \cdots (x + d).$$

Lemma 6. *Let*

$$0 \leq i < j \leq \left\lfloor \frac{n}{2} \right\rfloor, \quad d = j - i.$$

Then

$$\binom{n}{i} \mid \binom{n}{j} \iff A(i, d) \mid A(n - j, d).$$

Proof. We have

$$\frac{\binom{n}{j}}{\binom{n}{i}} = \frac{i!(n - i)!}{j!(n - j)!}.$$

Since $j = i + d$,

$$\frac{i!}{j!} = \frac{1}{(i + 1)(i + 2) \cdots (i + d)}$$

and

$$\frac{(n - i)!}{(n - j)!} = (n - j + 1)(n - j + 2) \cdots (n - i).$$

Therefore

$$\frac{\binom{n}{j}}{\binom{n}{i}} = \frac{A(n - j, d)}{A(i, d)}.$$

The quotient is an integer exactly when $A(i, d) \mid A(n - j, d)$. □

Example. Let $n = 10$, $i = 1$, and $j = 3$. Then

$$\binom{10}{1} = 10, \quad \binom{10}{3} = 120,$$

so $10 \mid 120$. Here $d = 2$, and the lemma replaces this by

$$A(1, 2) = 2 \cdot 3 = 6, \quad A(7, 2) = 8 \cdot 9 = 72.$$

Indeed, $6 \mid 72$, and the quotient is unchanged:

$$\frac{120}{10} = \frac{72}{6} = 12.$$

The lemma reduces the problem to divisibility between products of consecutive integers, allowing us to apply a theorem of Erdős and Straus.

4 The prime barrier

We use Theorem 3.3 of Erdős and Straus [3]. They write

$$A_{\text{ES}}(n, k) = (n+1)(n+2) \cdots (n+k),$$

which is exactly our function $A(n, k)$. Thus their theorem can be stated in our notation as follows.

Theorem 7 (Erdős–Straus, Theorem 3.3). *For every fixed $C > 1$, there exist only finitely many triples of positive integers (x, d, y) with*

$$d > 1, \quad x + d < y < Cx,$$

for which

$$A(x, d) \mid A(y, d)$$

and the interval

$$[x+1, x+d]$$

contains a prime.

For our application, Theorem 7 gives the following consequence.

The interpretation of the theorem of Erdős and Straus as a prime barrier for divisibility relations in rows of Pascal’s triangle, as well as the construction used in the proof of Theorem 1, were proposed by ChatGPT and subsequently verified by the author.

Corollary 8. *Let $0 < \alpha < 1/2$ be fixed. Then, for all sufficiently large n , if*

$$\alpha n \leq i < j < \frac{n}{2}, \quad j - i > 1,$$

and

$$\binom{n}{i} \mid \binom{n}{j},$$

then the interval $(i, j]$ contains no prime.

Proof. By Lemma 6,

$$A(i, j-i) \mid A(n-j, j-i).$$

Set

$$x = i, \quad d = j - i, \quad y = n - j.$$

Since $j - i > 1$, we have $d > 1$. For sufficiently large n we also have $x = i > 0$. Since $j < n/2$,

$$x + d = j < n - j = y.$$

Furthermore, $i \geq \alpha n$ gives

$$y < n \leq \frac{x}{\alpha}.$$

Thus $y < Cx$ with the fixed constant $C = 1/\alpha$. All hypotheses of Theorem 7 are therefore satisfied. Its exceptional cases are finite in number, while

$$x = i \geq \alpha n \rightarrow \infty.$$

Consequently, for all sufficiently large n the interval

$$[x + 1, x + d] = (i, j]$$

contains no prime. □

Under the assumptions of Corollary 8, every prime between two positions acts as a barrier against non-neighboring divisibility.

5 From the prime barrier to large divisibility-free sets

Proof of Theorem 1. Fix $0 < \alpha < 1/2$. For every prime q with

$$\alpha n < q < \frac{n}{2} - 1,$$

select the two positions q and $q + 1$.

For sufficiently large n , all such primes are odd. Hence two distinct selected primes differ by at least 2, so the pairs $\{q, q + 1\}$ do not overlap.

Let $q < r$ be two distinct selected primes, and let

$$i \in \{q, q + 1\}, \quad j \in \{r, r + 1\}.$$

If i and j are not neighboring, then

$$i \leq q + 1 < r \leq j,$$

so the prime r lies in $(i, j]$. By Corollary 8, for sufficiently large n ,

$$\binom{n}{i} \nmid \binom{n}{j}.$$

The reverse divisibility is impossible because the binomial coefficients increase strictly up to the middle of the row.

It remains to consider neighboring selected positions. By Proposition 3,

$$\binom{n}{k} \mid \binom{n}{k+1} \iff k+1 \mid n+1.$$

Every problematic neighboring pair therefore corresponds to a divisor $d = k + 1 > \alpha n$ of $n + 1$. Its complementary divisor satisfies

$$\frac{n+1}{d} < \frac{n+1}{\alpha n}.$$

For fixed α the right-hand side is bounded, so there are only $O_\alpha(1)$ possible complementary divisors and hence only $O_\alpha(1)$ problematic neighboring pairs.

Removing at most one position from each such pair costs only $O_\alpha(1)$ elements. The number of primes used is

$$\pi\left(\frac{n}{2}\right) - \pi(\alpha n) + O(1),$$

so at least

$$2\left(\pi\left(\frac{n}{2}\right) - \pi(\alpha n)\right) - O_\alpha(1)$$

positions remain. By the Prime Number Theorem,

$$\pi(x) \sim \frac{x}{\log x},$$

and hence

$$a(n) \geq (1 - 2\alpha + o(1)) \frac{n}{\log n}.$$

Since $\alpha > 0$ can be chosen arbitrarily small,

$$\liminf_{n \rightarrow \infty} \frac{a(n) \log n}{n} \geq 1,$$

which is equivalent to

$$a(n) \geq (1 - o(1)) \frac{n}{\log n}.$$

□

A small example. Let $n = 31$ and $\alpha = 1/5$. The primes in

$$\alpha n < q < \frac{n}{2} - 1$$

are 7, 11, 13, so we first select

$$7, 8, 11, 12, 13, 14.$$

Since $n + 1 = 32$ and $8 \mid 32$, Proposition 3 gives

$$\binom{31}{7} \mid \binom{31}{8}.$$

We may remove position 7. For this specific small value one can check directly that no further divisibility occurs among

$$8, 11, 12, 13, 14.$$

This example is not an application of the asymptotic Corollary 8 to $n = 31$; it only illustrates the mechanism of the construction.

6 Why rows of the form $n = p - 1$ are especially favorable

For general rows, Theorem 1 gives a divisibility-free set of order $n/\log n$. For the special family $n = p - 1$, where p is an odd prime, an additional congruence strengthens the prime barrier:

$$\binom{p-1}{k} \equiv (-1)^k \pmod{p}.$$

Indeed,

$$\binom{p-1}{k} = \frac{(p-1)(p-2)\cdots(p-k)}{k!},$$

and $p - r \equiv -r \pmod{p}$.

6.1 From the congruence to a large quotient

Assume that

$$0 \leq i < j \leq \frac{p-1}{2}$$

and

$$\binom{p-1}{i} \mid \binom{p-1}{j}.$$

Since the binomial coefficients increase strictly up to the middle of the row,

$$R = \frac{\binom{p-1}{j}}{\binom{p-1}{i}}$$

is an integer greater than 1. The congruence gives

$$R \equiv (-1)^{j-i} \pmod{p}.$$

Thus $R \equiv \pm 1 \pmod{p}$, and since $R > 1$,

$$R \geq p - 1.$$

Every nontrivial divisibility relation therefore produces a large quotient. However, throughout the range

$$\frac{p-1}{4} \leq i < j \leq \frac{p-1}{2},$$

this quotient can grow only slowly.

6.2 A logarithmic minimum separation

The use of the congruence $\binom{p-1}{k} \equiv (-1)^k \pmod{p}$ to derive a logarithmic minimum separation between comparable coefficients was proposed by ChatGPT as a proof strategy.

Lemma 9. *Let*

$$\frac{p-1}{4} \leq i < j \leq \frac{p-1}{2}$$

and suppose

$$\binom{p-1}{i} \mid \binom{p-1}{j}.$$

Then

$$j - i > \log_3(p-1).$$

Proof. As above,

$$R = \frac{\binom{p-1}{j}}{\binom{p-1}{i}} \geq p-1.$$

On the other hand,

$$R = \prod_{r=i}^{j-1} \frac{p-1-r}{r+1}.$$

For $r \geq (p-1)/4$,

$$p-1-r \leq \frac{3(p-1)}{4}, \quad r+1 > \frac{p-1}{4},$$

so every factor is less than 3. Hence

$$R < 3^{j-i}.$$

Combining the inequalities,

$$p-1 \leq R < 3^{j-i},$$

and therefore

$$j - i > \log_3(p-1).$$

□

Example. For $p = 31$,

$$\log_3(30) = 3.09\dots$$

Thus, among the integer positions $8, \dots, 15$, two coefficients that are comparable under divisibility must be at least 4 positions apart. In particular, divisibility between positions 10 and 12, or between 11 and 14, is impossible. The lemma gives only a necessary condition; a large distance does not guarantee divisibility.

6.3 Neighboring divisibility disappears

In the same index range, neighboring divisibility is impossible. By Proposition 3,

$$\binom{p-1}{k} \mid \binom{p-1}{k+1}$$

could hold only if $k+1 \mid p$. Since p is prime and $1 < k+1 < p$ in the range under consideration, this cannot happen.

Thus, in the index interval from $(p-1)/4$ to $(p-1)/2$,

- neighboring coefficients do not divide each other;
- comparable non-neighboring coefficients must be more than $\log_3(p-1)$ positions apart.

Since the binomial coefficients increase strictly up to the middle, reverse divisibility is also impossible. Consequently, sufficiently short intervals in this region are automatically divisibility-free.

7 Prime gaps as building blocks

Let $X > 0$ and $K > 0$. We denote by

$$A_1(K; X)$$

the set of primes $q \leq X$ such that $(q, q + K]$ contains no other prime. Thus $q \in A_1(K; X)$ is the starting point of a prime gap of length at least K .

For example, there is no prime between 31 and 37, so

$$31 \in A_1(5; 50),$$

because $(31, 36]$ is prime-free.

For our application we set

$$K = \lambda \log X,$$

where $\lambda > 0$ is fixed. We do not need exceptionally large prime gaps; logarithmically long gaps are sufficient, provided they occur often enough.

Theorem 10 (Bazzanella–Languasco–Zaccagnini, Theorem 5). *Let $\varepsilon > 0$ be fixed. For sufficiently large X and*

$$0 < \lambda < \frac{2}{B} - \varepsilon,$$

we have

$$|A_1(\lambda \log X; X)| \geq \left(1 - \frac{\lambda(B + \varepsilon)}{2}\right) \frac{X}{\log X},$$

where one may take $B = 3.454$.

This is Theorem 5 of Bazzanella, Languasco, and Zaccagnini [1]. They write the set as $A_1(K)$ because X is fixed throughout their discussion; we write $A_1(K; X)$ to make the dependence on X explicit.

For every sufficiently small fixed λ , the theorem therefore supplies a number of suitable primes of order $X/\log X$, each followed by a prime-free interval of length about $\lambda \log X$.

Set

$$h = \lfloor \lambda \log X \rfloor.$$

Each such prime q gives a block

$$I_q = \{q, q + 1, \dots, q + h\}.$$

For sufficiently small λ , these blocks are shorter than the minimum separation required by Lemma 9.

8 Many short blocks give a linear-sized set

The connection between the logarithmic minimum separation and the result of Bazzanella, Languasco, and Zaccagnini on logarithmic prime gaps was proposed by ChatGPT.

Proof of Theorem 2. Fix

$$0 < \lambda < \frac{1}{B}, \quad B = 3.454,$$

and set

$$X = \frac{p-1}{2}, \quad H = \lfloor \lambda \log X \rfloor.$$

For each prime q with

$$\frac{X}{2} < q \leq X - H - 1,$$

let q^+ be the next prime after q and write

$$g_q = q^+ - q.$$

Instead of using only prime gaps of length at least $H + 1$, we consider the block

$$J_q = \{q, q+1, \dots, q + \min(H, g_q - 1)\}.$$

Thus every such prime gap contributes, possibly with a shorter block.

All positions in these blocks lie in

$$\frac{p-1}{4} < i < \frac{p-1}{2}.$$

Each block J_q is divisibility-free. Indeed, its diameter is at most H . Since $\lambda < 1/B < 1/\log 3$, we have

$$H < \log_3(p-1)$$

for all sufficiently large p . By Lemma 9, divisibility between nonadjacent indices in the same block is therefore impossible. Divisibility between adjacent indices was excluded earlier, and the reverse divisibility cannot occur because the binomial coefficients increase strictly up to the middle of the row.

We claim that the union of all blocks J_q is an antichain. Let $q < r$ be two starting primes. Since $q^+ \leq r$, the block J_q ends no later than $q^+ - 1$. Hence every $i \in J_q$ satisfies $i < r$, while every $j \in J_r$ satisfies $r \leq j$. Therefore

$$i < r \leq j.$$

The prime r lies in the interval $(i, j]$. If $j - i > 1$, Corollary 8 applies. If $j - i = 1$, divisibility is excluded by Proposition 3. Thus no divisibility relation can occur between different blocks, and the union of all J_q is an antichain.

For fixed $t \in (0, \lambda]$, define

$$N_X(t) = \#\{q : X/2 < q \leq X - H - 1, \text{ } q \text{ prime, } g_q > t \log X\}.$$

By Theorem 10 and the Prime Number Theorem, after subtracting the primes up to $X/2$ we obtain the lower bound below. The additional upper cutoff $q \leq X - H - 1$ removes at most $O(\log X)$ possible primes and does not affect the asymptotics. Letting ε tend to 0, we obtain

$$\liminf_{X \rightarrow \infty} \frac{N_X(t) \log X}{X} \geq \frac{1 - Bt}{2}.$$

Here t is fixed before taking the limit $X \rightarrow \infty$, so no uniformity in t is required.

Let S_X denote the total number of selected indices. Since

$$|J_q| = \min(H + 1, g_q),$$

we obtain, for every integer $m \geq 1$, by setting $t_r = r\lambda/m$,

$$S_X \geq \log X \sum_{r=1}^m (t_r - t_{r-1}) N_X(t_r).$$

For fixed m , the number of thresholds is finite, so

$$\liminf_{X \rightarrow \infty} \frac{S_X}{X} \geq \sum_{r=1}^m (t_r - t_{r-1}) \frac{1 - Bt_r}{2}.$$

Letting $m \rightarrow \infty$ gives

$$\liminf_{X \rightarrow \infty} \frac{S_X}{X} \geq \int_0^\lambda \frac{1 - Bt}{2} dt = \frac{\lambda}{2} - \frac{B\lambda^2}{4}.$$

Because $p = 2X + 1$, we obtain

$$\liminf_{\substack{p \rightarrow \infty \\ p \text{ prime}}} \frac{a(p-1)}{p} \geq \frac{\lambda}{4} - \frac{B\lambda^2}{8}.$$

□

8.1 An explicit constant

The right-hand side is increasing for $0 < \lambda < 1/B$. Letting λ tend to $1/B$ from below yields

$$\liminf_{\substack{p \rightarrow \infty \\ p \text{ prime}}} \frac{a(p-1)}{p} \geq \frac{1}{8B}.$$

With $B = 3.454$, this gives

$$\frac{1}{8B} = 0.03618992 \dots$$

Thus the bound is linear along $n = p - 1$, in contrast with the general lower bound of order $n/\log n$.

9 Exact computation of the sequence values

The formulation of $a(n)$ as the width of the divisibility poset, and the reduction of its exact computation to Dilworth's theorem and a maximum matching in a bipartite graph, were proposed during the AI-assisted research process with ChatGPT and subsequently verified by the author.

For a fixed n , we would also like to determine $a(n)$ exactly. We equip P_n with the divisibility order

$$x \preceq y \iff x \mid y.$$

Then a divisibility-free subset is exactly an antichain, so $a(n)$ is the width of this poset.

Theorem 11 (Dilworth). *In every finite partially ordered set, the maximum size of an antichain equals the minimum number of chains needed to cover all elements.*

Thus $a(n)$ is the minimum number of chains in a chain cover of P_n . As in the standard bipartite-graph proof of Dilworth's theorem, this minimum can be computed by a maximum matching.

Set

$$m = \left\lfloor \frac{n}{2} \right\rfloor + 1.$$

Construct a bipartite graph G_n with a left and a right copy of the indices $0, 1, \dots, m-1$. Connect the left copy of i to the right copy of j whenever

$$i < j \quad \text{and} \quad \binom{n}{i} \mid \binom{n}{j}.$$

Let $\nu(G_n)$ be the size of a maximum matching.

Suppose a chain cover has c chains containing all m elements. Within each chain, connect consecutive elements. Since the binomial coefficients increase strictly up to the middle of the row, comparable elements in a chain correspond to increasing indices. This produces exactly $m - c$ edges, and these edges form a matching in the bipartite graph. Hence

$$\nu(G_n) \geq m - c.$$

For the minimum chain number $c_{\min}$,

$$c_{\min} \geq m - \nu(G_n).$$

Conversely, a matching with $\nu(G_n)$ edges gives directed links from smaller to larger comparable elements. Each index is used at most once as a left endpoint and at most once as a right endpoint, and the order contains no directed cycles. The links therefore decompose into disjoint directed paths. By transitivity of divisibility, every such path is a chain. Thus the matching yields a chain cover with

$$m - \nu(G_n)$$

chains, so

$$c_{\min} \leq m - \nu(G_n).$$

Therefore

$$c_{\min} = m - \nu(G_n).$$

By Dilworth's theorem, $c_{\min} = a(n)$, and hence

$$a(n) = m - \nu(G_n).$$

The first values of [A399060](#), $a(0), \dots, a(100)$, are

1, 1, 1, 1, 2, 1, 3, 2, 3, 3, 4, 2, 4, 4, 4, 4, 7, 5, 6, 5, 5, 7, 8, 6, 8, 8, 8, 9, 10, 8, 11, 8, 10, 11, 10, 9, 13, 14, 13, 12, 14, 12, 15, 13, 14, 16, 18, 15, 16, 16, 17, 19, 19, 16, 17, 16, 20, 22, 24, 20, 24, 23, 22, 21, 21, 20, 23, 24, 22, 23, 28, 22, 28, 26, 24, 24, 26, 25, 29, 26, 27, 31, 32, 30, 32, 33, 32, 31, 34, 30, 33, 35, 35, 38, 36, 32, 40, 39, 36, 38, 41.

10 Open questions

The two main theorems provide general asymptotic lower bounds for $a(n)$, but a large gap remains. We have

$$a(n) \geq (1 - o(1)) \frac{n}{\log n},$$

whereas for $n \geq 2$ the trivial upper bound

$$a(n) \leq \left\lfloor \frac{n}{2} \right\rfloor$$

holds, since $\binom{n}{0} = 1$ divides every other entry.

Open Question 1. Does there exist an absolute constant $c > 0$ such that

$$a(n) \geq cn$$

for all sufficiently large n ?

Theorem 2 shows that linear growth does occur along the infinite subsequence $n = p - 1$.

Open Question 2. Is it true that, as $p \rightarrow \infty$ through the primes,

$$a(p - 1) = \frac{p - 1}{2} - o(p)?$$

If so, asymptotically almost the entire set of available distinct entries could be selected; divisibility would exclude only a vanishing fraction of the positions.

The present methods are not strong enough to prove this. Corollary 8 indicates a major difficulty: in the middle region covered by that corollary, divisibility is asymptotically strongly restricted whenever a prime lies between two positions. The remaining difficult cases are therefore concentrated in longer prime-free intervals.

This leads to a more local question: which divisibility relations can actually occur inside a long prime-free interval? A better understanding of this structure could lead both to stronger lower bounds for general rows and to a substantial improvement of Theorem 2.

Statement on the Use of Artificial Intelligence

This project was developed through an AI-assisted research process using ChatGPT by OpenAI.

ChatGPT played a substantial role in the mathematical development of the paper. The main ideas, proof strategies, and first versions of the proofs were proposed by ChatGPT. The specific contributions are explicitly acknowledged at the corresponding places in the paper. I directed the research process, critically evaluated the proposed approaches, and decided which arguments should be pursued, modified, or discarded. All mathematical statements, proofs, calculations, and conclusions included in the final version were carefully checked and verified by me. I take full responsibility for the mathematical correctness of the paper.

ChatGPT was also used to search for potentially relevant mathematical literature. Every reference used in the final paper was checked by me against the original source. In particular, I compared the results of Erdős and Straus and of Bazzanella, Languasco, and Zaccagnini used in the paper directly with the original publications and independently verified the assumptions required for their application.

The mathematical working version was initially prepared in German. After this version had been completed and verified, I translated the manuscript into English myself. AI tools were used during the preparation of the English version only to identify possible issues with spelling, grammar, mathematical terminology, consistency, and idiomatic language. The final English wording of the submitted version was chosen and written by me. The submitted English version was not translated or linguistically rewritten by a large language model.

ChatGPT was also used for the technical preparation and formatting of the LaTeX manuscript and for checking compliance with the formal requirements of the Journal of Integer Sequences. This technical assistance did not alter either the English wording written by the author or the mathematical content of the manuscript.

References

- [1] D. Bazzanella, A. Languasco, and A. Zaccagnini, Prime numbers in logarithmic intervals, *Trans. Amer. Math. Soc.* **362** (2010), 2667–2684.
- [2] R. P. Dilworth, A decomposition theorem for partially ordered sets, *Ann. of Math. (2)* **51** (1950), 161–166.
- [3] P. Erdős and E. G. Straus, On products of consecutive integers, in H. Zassenhaus, ed., *Number Theory and Algebra*, Academic Press, 1977, pp. 63–70.
- [4] H. Joris, C. Oestreicher, and J. Steinig, The greatest common divisor of certain sets of binomial coefficients, *J. Number Theory* **21** (1985), 101–119.
- [5] OEIS Foundation Inc., *A399060: Maximum size of a subset of the distinct entries in row n of Pascal's triangle such that no selected entry divides another*, The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A399060>, 2026.
- [6] D. Singmaster, Divisibility of binomial and multinomial coefficients by primes and prime powers, in V. E. Hoggatt, Jr. and M. Bicknell-Johnson, eds., *A Collection of Manuscripts Related to the Fibonacci Sequence: 18th Anniversary Volume*, Fibonacci Association, 1980, pp. 98–113.

(Concerned with sequence [A399060](https://oeis.org/A399060).)